A Living Atlas of Recorded Color

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[FinalProject4: Spatial Color Atlas] | [Andi/aw565 & Tingyi / tx50 & Jiaxin Wang/jw848] | Version or commit: 4368799

Abstract: A Living Atlas of Recorded Color is a source-linked color atlas for comparing sampled palettes across places in China. It is designed for design/media students, creative practitioners, and cultural educators whose task is to explore a place, inspect recorded palettes and source photographs, compare samples, and trace similarities or differences back to photographic evidence. The verified dataset contains 50 team-authored photographs from five places. Five linked representations—basic palette views, a similarity network, a geographic atlas, source-linked interaction, and an advanced draggable Compare view—serve this task, while Figures 1–3 document the views, system architecture, and evidence-based revision path. A research-informed pipeline uses deterministic K-means++ in CIELAB and CIEDE2000 matching. Peer feedback prompted richer comparison idioms, algorithm clarification, and the advanced effect. The project is intended to support provenance-aware creative reference and aligns with SDG 11.4 without claiming demonstrated heritage impact. Next validation will expand and rebalance the dataset and prototype real-time user photo upload, with optional consent-based contribution to a larger dataset.

Keywords: interactive data visualization; cultural color atlas; Human-Centered Spatiotemporal Intelligence; DKU creative community; SDG 11.4

1 Project Frame and Contribution

1.1 Question, user, task, and local context

Our project asks how an interactive, source-linked visualization can help users discover similarities and differences among sampled cultural color palettes from different places in China without reducing those places to fixed or representative color identities. The intended users are design and media students, creative practitioners, and cultural educators who may use visual references from different places as inspiration for creative or educational work.

The exact task is to explore a place, inspect its recorded palette and source photographs, compare it with another sampled place, and trace similarities or differences back to the photographic evidence from which they were derived. The current local audience is primarily students and educators in the DKU community who work with visual, cultural, or media material. Our current need claim is therefore a hypothesis: the atlas may make place-based visual references easier to explore while keeping their provenance visible, but we have not yet established broader educational or creative benefit.

The project grew from an earlier Jinxi pilot, where the team observed color variation across water, architecture, materials, and everyday scenes. That pilot also revealed an important limitation: a short photographic visit can reveal patterns worth comparing but cannot establish a culturally recognized or representative palette. The expanded project therefore treats all five locations as sampled photographic records rather than authoritative cultural descriptions. Residents, cultural practitioners, and heritage experts from the sampled places remain missing voices. The current photographs are team-authored and contain no known identifiable people or private material requiring additional permission, although attribution, metadata, and culturally sensitive content should still be reviewed before broader public release.

1.2 Evidence-supported claim and boundary

The current artifact supports the following claim: Users can inspect and compare color patterns extracted from the current photographic samples while tracing aggregate palettes and similarity results back to source images.

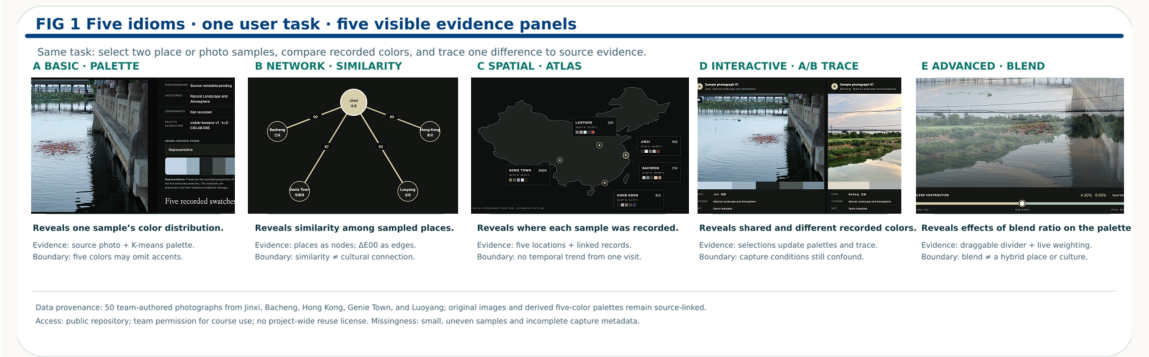
The evidence does not establish that these palettes are the true, typical, historically stable, or culturally recognized colors of Jinxi, Bacheng, Hong Kong, Genie Town, Luoyang, or China more broadly. Lighting, weather, camera settings, route selection, photographer choices, time of day, season, and uneven photographic coverage may all affect the recorded results. The project therefore uses terms such as recorded colors, sampled photographs, recorded palette**,** and sampled palette similarity, rather than “the colors of” a place.

1.3 Project contribution

The design contribution is a source-linked color atlas that connects computational color analysis with photographic and spatial context. Instead of separating palette extraction from its evidence, the interface lets users move between place-level patterns, comparisons, similarity relationships, scene categories, and individual photographs.

The primary emerging frontier is Human-Centered Spatiotemporal Intelligence, because the project connects computationally derived color information to geographically and culturally situated places while making missing context and sampling uncertainty visible. The project is aligned with SDG 11: Sustainable Cities and Communities, Target 11.4, through an intended contribution to documenting and communicating place-based cultural material; this is curricular alignment rather than demonstrated heritage impact. The course materials explicitly identify SDG 11.4 as appropriate where projects document community context, representation risks, permission, and routes for local validation.

Figure 1. Five representations serving one user task.



Representation	Task served	Input or interaction	Evidence and boundary
Basic idiom	Inspect/compare palettes	Ranking, photo detail, or A/B	Current samples only
Network idiom	Find palette relations	Select place; inspect links	Similarity represents palette similarity, not cultural similarity
Spatial idiom	Locate samples	Select mapped place	Uneven coverage; not representative
Interactive view	Trace aggregate to source	Browse, compare, inspect, reset	Traceability and recovery
Advanced representation	Inspect subtle A/B differences	Drag divider; palette updates	Exploratory samples; no cultural/temporal claim

2 Design Capability

2.1 Design pipeline

The domain problem is the difficulty of comparing place-based color references without turning limited photographic samples into claims about cultural identity. The data consist of team-produced photographs associated with places and scene categories, together with derived dominant colors and proportions. The primary task is to compare sampled palettes while remaining able to trace each result to its source photographs.

The visual idioms address different parts of the same task. Palette rankings support direct comparison; the network represents sampled-palette similarity; detail views preserve source context; and the geographic atlas situates recorded samples. The advanced Compare interaction lets users select two photographs and drag a divider to change their visible/blended ratio while the weighted blend palette updates continuously.

The implemented algorithm uses perceptual rather than raw RGB distance. Each source JPEG is orientation-corrected, resized, converted from sRGB to CIELAB D65, and processed with deterministic K-means++ (k=5) to obtain dominant colors and pixel-area proportions. Place palettes give each photograph equal total weight before weighted aggregation. Palette comparison uses optimal one-to-one matching and CIEDE2000 (ΔE00); proportion-weighted distances become a 0–100 similarity index. The method is informed by Celebi (2011), Gramazio et al. (2017), and Westland et al. (2024), and checked against Sharma et al.'s standard CIEDE2000 data. Similarity describes sampled photographic palettes, not cultural similarity.

Table 1. End-to-end design pipeline.

Domain	Data and task	Idiom	Algorithm
Explore culturally situated recorded colors without treating samples as place identity. Users are design and media students, creative practitioners, and cultural educators	Team photographs, place/category metadata, dominant colors and proportions; inspect, compare, trace	Palette ranking, A/B comparison, similarity network, geographic atlas, source-linked photo detail, and advanced Compare interaction	JPEG → sRGB/CIELAB D65 → deterministic K-means++ (k=5) → colors/proportions → equal-photo-weight aggregation → optimal palette matching → CIEDE2000 → weighted distance → 0–100 similarity

2.2 Four validation levels

Table 2. Four-level validation summary.

Level	Current evidence	Main threat	Next test and owner
Domain situation	Earlier Jinxi fieldwork and team discussion suggest value in comparing recorded place-based colors, but the broader creative/educational need remains a hypothesis	Current evidence still comes mainly from the project team and student proxies rather than a broader range of intended users or people familiar with the represented places.	Interview design/media students and cultural educators to validate the creative/educational need, and consult residents or cultural practitioners familiar with the sampled places to challenge overgeneralized cultural interpretations. Owner: Interaction & Evaluation.
Data and task abstraction	Five locations and source photographs are represented; palette results remain source-linked	Small, uneven photographic samples may overrepresent route, lighting, weather, or photographer choice	Expand the dataset using more balanced coverage across places and scene categories, and document relevant sampling metadata such as time, weather, route, and photographer where available. Owner: Data Visualization & Prototyping.
Visual encoding and interaction idiom	Current MVP supports place browsing, A/B comparison, categories, and similarity exploration	The A/B comparison workflow for the advanced effect part is not sufficiently self-explanatory; users may hesitate over where to select, the A, B photos since the selection is on the comparison page, while advanced effect is on the homepage.	Add concise in-context instructions and clearer selection-state feedback, then repeat the comparison task with a new user to test whether it can be completed without verbal guidance. Owner: Interaction & Evaluation.
Algorithm	The implemented pipeline reproducibly extracts five-color palettes from all 50 photographs, aggregates place palettes with equal photo weighting, computes CIEDE2000-based optimal matches, and passes standard and consistency checks.	k=5 is lossy and may omit small but culturally meaningful accents; rankings may depend on extraction, weighting, aggregation, and score-conversion choices.	Run parameter-sensitivity tests; compare algorithmic rankings with human judgments; inspect whether salient accents are systematically missed. Owner: Data Visualization & Prototyping.

2.3 Implementation, architecture, accessibility, and fallback

The prototype is a browser-based, data-driven visualization whose place, photograph, category, palette, comparison, and similarity records are separated from interface components. Source JPEGs can be regenerated through scripts/extract_palettes.py, with parameters and file hashes recorded in data/extraction-manifest.json. The submitted artifact contains 50 photographs across five places and links derived results back to their sources.

Earlier critique identified a small back button and low-contrast dashed similarity links; both have now been revised through greater prominence and visual weight. The interface also provides clear/reset/return states. These fixes improve readability but do not constitute broad accessibility or usability validation. The current static application has no dedicated user-facing recovery mechanism for missing or malformed resources; for transient loading problems, users can refresh the page, while the documented local static-server workflow remains the operational fallback.

For the advanced representation, the simpler alternative is a static side-by-side photo/palette comparison. The implemented Compare interaction adds a draggable divider that continuously changes the visible/blended ratio of two source photographs while updating the weighted blend palette. This supports closer inspection of transitions between two samples, while static comparison remains the fallback.

Figure 2. Repository and software architecture for the submitted version.

Architecture record	Data and provenance	Transform and store	Logic and algorithms	Views and interaction	Deploy, test, and fallback
Repository path or service	images/; data/photos.json; data/places.json; data/categories.json; data/extraction-manifest.json	scripts/extract_palettes.py; requirements.txt; generated JSON records	app.js and relevant similarity logic	index.html; app.js; styles.css	README.md; GitHub Pages; local static server
Project-specific component	50 team-authored photographs; place/category/photo metadata; manifest parameters, versions, and SHA-256 records	EXIF correction; resize; sRGB → CIELAB D65; deterministic K-means++ (k=5); proportions; equal-photo-weight aggregation	Optimal one-to-one matching; CIEDE2000; proportion weighting; 0–100 similarity; blend-palette logic	Geographic atlas; place/category/photo views; A/B comparison; similarity network; source trace; Compare slider	Live artifact; reproducibility instructions; static browser app; local fallback
Input, output, and check	Input: JPEGs + metadata. Output: verified records. Check: provenance, hashes, sample boundaries, sharing review.	Input: JPEGs. Output: reproducible photo/place palettes. Check: fixed seed, manifest, regeneration consistency.	Input: palettes, proportions, state. Output: matches, weighted ΔE00, score, blend palette. Check: Sharma tests, self=100, symmetry, offline/browser consistency.	Input: selection/slider state. Output: map, palettes, source images, similarity, blend palette. Check: clear/reset/return and revised visibility.	Input: repository. Output: GitHub Pages site. Check: clean-browser/manual test; documented local-server fallback.

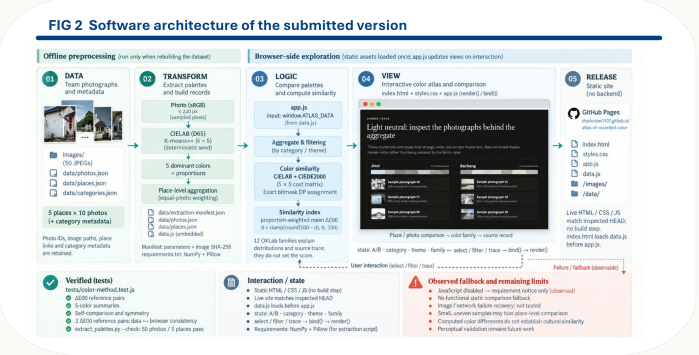


Figure description: 50 source JPEGs with place, category, and photo metadata are processed through a reproducible color-extraction pipeline, normalized and reduced to five dominant CIELAB colors using deterministic K-means++. Extracted palettes are compared using CIEDE2000 matching with proportion weighting to produce 0–100 similarity scores. The results are presented through the geographic atlas, source-linked views, A/B comparison, similarity network, and Compare interaction. GitHub Pages is the primary deployment, with a local static server as the documented fallback.

Artifact URL: https://charleston1107.github.io/atlas-of-recorded-color/ Repository URL: https://github.com/charleston1107/atlas-of-recorded-color

Version or commit: 4368799

Poster: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Poster.pdf

Final paper: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Report.pdf

Demo video: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_DemoVideo.mp4

3 Interdisciplinary Contribution

3.1 Three emerging frontiers

Table 3. Frontier-to-design contribution.

Emerging frontier	Contribution and evidence	Visible design consequence	Current limit
Open Science, Data Governance, and Responsible AI	Team-authored photographs; public repository; extraction script, requirements, manifest with SHA-256 records, generated data, and automated consistency checks	Source-linked records, reproducible transformations, explicit recorded/inferred distinction, and documented AI assistance	No explicit open-source license is assigned; public access does not imply unrestricted reuse. Public-sharing and cultural-context review remain limited.
Scientific Communication for Emerging Technologies	Classroom critique exposed weak algorithm explanation; the revision documents extraction/matching logic and preserves source-image traceability	Evidence-bounded terminology, visible method, sample limits, and source-linked palette results	Evidence comes from classroom peers and instructor feedback rather than a broader intended-user sample.
Human-Centered Spatiotemporal Intelligence (Primary)	Recorded palettes are linked to geographically situated photographs without claiming a fixed cultural identity	Implemented geographic atlas, source access, sample boundaries, and sampled-palette similarity framing	Weather, season, and time remain sampling limitations rather than the current analytic focus; broader local validation is absent.

3.2 Research-to-design change

Classroom critique identified that our original palette-similarity method lacked sufficient theoretical grounding and was difficult to explain. We therefore revised the algorithm and its documentation. The new pipeline uses deterministic K-means++ in CIELAB for color quantization, informed by Celebi (2011), and optimal palette matching with CIEDE2000, informed by Westland et al. (2024); its use of perceptual color distance is consistent with the rationale in Colorgorical (Gramazio et al., 2017), not an implementation of the Colorgorical algorithm. Sharma et al.'s standard test data validate the CIEDE2000 implementation. This change is visible in the extraction script, manifest, updated comparison logic, tests, and source-linked interface.

4 Community-Rooted Global Leadership

4.1 Local community service

The immediate audience is DKU design/media students and related creative or educational practitioners. The intended service is to give creators a source-linked, traceable visual reference that can support an intuitive and affective impression of place-based color and potentially inform later creative work. Classroom feedback suggests this framing is understandable, but does not establish educational or cultural benefit.

Photographs are team-produced, yet attribution, metadata, identifiable people, and culturally sensitive content still require review before wider release.

Residents, maintainers, and cultural practitioners are not partners or co-designers. A future public atlas should provide source-transparent documentation, a correction route, and a named maintenance owner rather than claim present community benefit. A future version may also allow users to upload photographs for real-time palette extraction and comparison and, with explicit consent, optionally contribute those photographs to a larger dataset; this feature is not yet implemented.

4.2 SDG goal and target

The project aligns with SDG 11: Sustainable Cities and Communities, Target 11.4, by exploring provenance-aware documentation of sampled place-based visual material. Its intended pathway is to give creators source-linked color references that may inform culturally situated creative work and, through those works, potentially communicate regional visual material to broader audiences. This is an intended contribution pathway, not demonstrated heritage impact.

Current evidence shows only that samples were produced, visualized, and compared; it does not demonstrate preservation, cultural learning, community benefit, broader cultural communication, or SDG impact.



Table 4. Local service and SDG accountability.

Accountability field	Current evidence	Limit or next check
Need hypothesis and missing voices	DKU design/media audience identified; initial feedback from design/media students and a cultural educator suggests the task and framing are understandable	Evidence is limited to classroom critique; residents, local cultural practitioners, and heritage experts remain missing voices
Permission, protection, and public-sharing boundary	Current photographs are team-produced	Audit identifiable people, metadata, attribution, and any restricted material before wider release
Reciprocal return and correction route	Planned public atlas, method documentation, source linkage, and a future user-upload workflow for real-time palette extraction/comparison	Add explicit feedback/correction mechanism and maintenance owner; prototype opt-in dataset contribution with clear consent
SDG 11: Sustainable Cities and Communities, Target 11.4	Project documents sampled place-based visual material	Use alignment language only; no evidence of preservation impact

5 Interactive Evaluation and Evidence-Based Revision

5.1 Activity 1 Interactive Demonstration

On September 24, peers began at the atlas, opened place/photo views, compared samples, and inspected similarity. The static views provided fallback. They identified a small back button and faint network links; both were revised. Seasonal, time-of-day, and weather comparison was marked Investigate because it exceeds the current cross-place scope.

The revision increased navigation prominence and network contrast while retaining temporal/environmental factors as visible sampling limits.

This evidence addresses interaction-idiom readability, not broad usability.

5.2 Activity 2 In-Class Peer Evaluation / Progress Critique

On September 29, design/media peers and the instructor reviewed the default atlas, comparison flow, similarity evidence, and claim boundaries. Two verbal and five Ed comments highlighted weak algorithm explanation, the missing advanced effect, and visibility issues. The session produced actionable revisions but was not a formal recruited study.

Observed strength was source-linked comparison; the priority was clearer algorithm/interaction evidence. The next question is whether new intended users can complete A/B selection and interpret similarity without verbal guidance.

5.3 Activity 3 Response to Peers

Priority 1—Revise: weak grounding led to the CIELAB/K-means++/CIEDE2000 pipeline, manifest, regeneration script, and tests. Owner: Data Visualization & Prototyping.

Priority 2—Revise: the missing advanced effect led to the draggable Compare overlay and updating blend palette. Owners: Prototyping / Interaction & Evaluation.

Priority 3—Revise: back-button prominence and similarity-link contrast were increased. Owners: Interaction & Evaluation / Prototyping.

Priority 4—Investigate: seasonal, time, and weather comparison remains a future sampling question. Owner: Team.

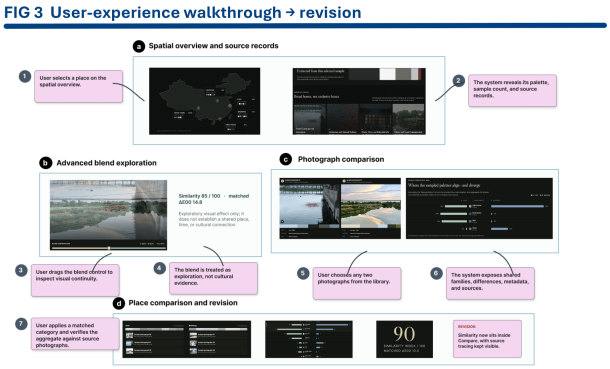


Figure 3. User-experience walkthrough and evidence-based revision.

Study record	User and exact task	Starting state	User action	System response and task result	Observed evidence	Feedback decision and revision
Task path	Design/media peer or classroom reviewer; inspect place palettes, compare samples, inspect similarity, and interpret evidence limits	Default atlas / homepage	Select place/photo; compare samples; inspect palette/similarity; use Compare slider where relevant	Displays source-linked palettes, similarity evidence, source images, and weighted blend palette	Critique identified weak algorithm explanation, missing advanced effect, and visibility problems; temporal suggestions raised a scope question	Revise algorithm; Revise advanced representation; Revise visibility; Investigate temporal feature
Evidence boundary	Classroom peer/instructor critique; no formal recruitment or recorded personal identifiers	Course presentation and prototype exploration	Feedback on design, scope, explanation, and interaction	Resulted in algorithm, Compare, documentation, and UI revisions	Untested: broad usability, learning, long-term creative benefit, cultural validity, community benefit, and SDG impact	Next: broader intended-user evaluation plus algorithm/human-similarity validation

Figure description: Classroom reviewers explored the source-linked atlas and comparison flow. Their critique identified algorithm-explanation, required-effect, and visibility issues; the submitted artifact responds through a reproducible color pipeline, the Compare interaction, and revised navigation/network contrast.

5.4 What the evidence does not establish

This peer critique cannot establish general usability, learning, creative or community benefit, cultural validity, or SDG impact.

6 Results, Limitations, and Next Validation

Table 5. Current result and next validation.

Claim or task	Current evidence	Main limitation	Next test and owner
Users can navigate sampled palettes, compare samples, and trace results to source photographs	Implemented place/photo browsing, comparison, similarity, advanced Compare, and revised navigation/link visibility; classroom peers inspected and critiqued the prototype	Evidence is classroom critique rather than broader task-based usability testing	Expand and rebalance photographic coverage; prototype user photo upload for real-time palette extraction/comparison. Owner: Team.
The algorithm can reproducibly generate and compare five-color sampled palettes	50-image regeneration, fixed-seed K-means++, CIEDE2000 tests, browser/offline consistency, self=100, and A↔B symmetry	k=5 may omit culturally salient accents; results may depend on parameter and weighting choices	Evaluate the extraction/comparison pipeline on a larger, more balanced dataset and uploaded-photo workflow. Owner: Data Visualization & Prototyping.

7 Bounded Contribution and Current Conclusion

The defensible contribution is a source-linked visualization architecture that combines geographic context, interactive comparison, a similarity network, and reproducible perceptual-color analysis while keeping evidence boundaries visible.

The interdisciplinary contribution connects open-science governance, scientific communication, and Human-Centered Spatiotemporal Intelligence; intended DKU/SDG 11.4 benefits remain hypotheses.

Next validation will expand and rebalance photographic coverage and prototype user photo upload for real-time palette extraction and comparison. With explicit consent, uploaded photographs may optionally contribute to a larger dataset. Five-color summaries may omit salient accents, and sampled-palette similarity is not cultural similarity.

Acknowledgements

We thank Prof. Luyao Zhang for instruction and guidance in INFOSCI 301. Michelle Clark provided early feedback encouraging richer visualization idioms rather than relying on text-heavy place comparison; this led us to expand the comparison with additional basic visual encodings. Yiran Zheng challenged whether the advanced draggable split view was more useful than static side-by-side comparison; we retained static comparison for detailed reading while positioning the draggable split view as a more engaging advanced/cover interaction. During the Week 7 duet defense, Jiajun Fu, Ruojin Wang, and Enwei Fan asked us to clarify the project motivation and its specific community contribution. We defended the atlas as a source-linked visual reference that can help creators form intuitive impressions of place-based color, while broader cultural communication and SDG 11.4 relevance remain intended rather than demonstrated outcomes; this clarification did not require an artifact change. We acknowledge course visits to Tencent's Shanghai Office, Shanghai Science and Technology Museum, Jinxi Intangible Cultural Heritage Museum, and China Ancient Brick and Tile Museum; these visits do not imply endorsement, partnership, or co-design. Observations of Suzhou water towns, including Jinxi and Bacheng, and original photographs taken by Andi Wan, Tingyi Xiao, and Jiaxin Wang informed the project with the photographers' permission.

Who / source	Exact contribution	Resulting change, attribution, or permission
Michelle Clark	Early feedback to add richer visualization idioms rather than rely on text-heavy place comparison.	Expanded the comparison with additional basic visual encodings.
Yiran Zheng	Asked whether the draggable split view was more useful than static side-by-side comparison.	Retained static comparison for detailed reading and used the draggable split view as the advanced/cover interaction.
Jiajun Fu, Ruojin Wang, Enwei Fan	Week 7 duet question: clarify project motivation and the specific community contribution.	Keep / Clarify: strengthened the final communication of the source-linked creator reference and bounded SDG 11.4 pathway; no artifact change from the defense.
Course field visits and team photographs	Observation and team-authored visual material; no partnership or co-design implied.	Credited as contextual observation and original photography; broader public-sharing review remains required.

AI and human responsibility

AI and human responsibility: ChatGPT 5.6 and Codex 5.6-Sol were used for project framing, language drafting, document revision, code/debugging support, and documentation. The team reviewed, tested, and revised all accepted outputs and retains responsibility for the submitted claims, design decisions, code, and evidence

Main-report preflight. Final submission checked for figure legibility, SDG attribution, acknowledgements, direct links, and version consistency.

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SIGCHI Guide for Videos. https://sigchi.org/resources/guides-for-authors/videos/

Appendix A Evidence Map and Version History

The Week 5 baseline is preserved so later revisions can be traced without overwriting prior evidence.

Table A1. Evidence and revision history.

Stage	Version or evidence	Change and reason	Current location
Tuesday exit gate	Week 5 planning / validation worksheet	Confirmed project/data thread, roles, MVP scope, and exact task	Week 5 worksheet / team records
Interactive Demonstration	Sep 24 MVP demonstration	Back button and dashed similarity links — Revise, now implemented; temporal palettes — Investigate	Section 5; Appendix D; commit ebe8720
Week 5 first attempt	Five-place atlas baseline	Source-linked browsing, A/B comparison, sampled-palette similarity, and evidence-bounded terminology	Artifact and repository baseline
Peer Evaluation	Sep 29 in-class peer evaluation / Ed Discussion feedback	Identified need for algorithm grounding, required advanced effect, clearer interaction explanation; temporal variation retained as scope limitation	Section 5; Appendices C/D; artifact/repository
Response to Peers	Post-feedback revision	Implemented extraction/similarity pipeline, advanced Compare effect, reproducibility records, and visibility improvements	Current artifact; commit 754758f
Week 7 → final submission	Oct 13 duet defense → final communication revision	Clarified motivation, intended creator/community contribution, bounded SDG 11.4 pathway, reviewer attribution, and final-submission evidence requirements; no artifact change from the duet defense.	Sections 1/4/7; Acknowledgements; Appendix D/E; 4368799

Table A2. Four validation levels.

Level	Current evidence	Main threat	Next test and owner
Domain situation	Five-place artifact; classroom critique from design/media peers and instructor	Benefit remains a hypothesis; residents and cultural stakeholders are absent	Broader intended-user/contextual validation. Owner: Interaction & Evaluation.
Data and task abstraction	50 team photographs from five places; palettes remain source-linked	Small/uneven samples; route, light, weather, season, time, and photographer bias	Coverage/metadata audit and balanced sampling. Owner: Data Visualization & Prototyping.
Visual encoding and interaction idiom	Geographic atlas, source views, A/B comparison, similarity network, and Compare	Compare guidance/discoverability lacks broader task testing	Add in-context guidance; test with intended users. Owners: Interaction & Evaluation + Data Visualization & Prototyping.
Algorithm	Reproducible CIELAB + fixed-seed K-means++ (k=5), equal-photo aggregation, matching, CIEDE2000, and tests	k=5 may miss accents; rankings depend on parameters and weights	Parameter sensitivity + human ranking comparison. Owner: Data Visualization & Prototyping.

Appendix B Data Governance and Reproducibility

The records below distinguish verified provenance from items requiring repository or team confirmation and state the project's prohibited inferences.

Table B1. Dataset and governance record.

Data record	Exact source and use	Risk or restriction	Verification
Primary dataset	50 team-authored photographs from five places; source-linked palette extraction, comparison, and browsing	Small/uneven samples and photographic-condition bias; cannot establish representative place colors	Ownership, metadata, hashes, and public-sharing status reviewed by the team
Complementary data	None used in the submitted analysis	No external dataset broadens or validates the photographic sample	Confirmed in submitted data and repository records
Synthetic or interface-test data	No synthetic dataset is used; automated tests use deterministic checks and standard CIEDE2000 reference values	Test values validate implementation, not cultural meaning or human similarity judgments	Sharma reference cases; reproducibility and consistency checks
Images, stories, or cultural material	Team-authored photographs and derived palettes	Extracted color does not establish cultural meaning; public access does not imply unrestricted reuse	Source linkage and public-sharing review; contextual interpretation remains unvalidated

Table B2. Reproducibility and release record.

Field	Record
Repository and artifact URLs	Artifact: https://charleston1107.github.io/atlas-of-recorded-color/ Repository: https://github.com/charleston1107/atlas-of-recorded-color
Tagged version or commit	4368799
Repository and software architecture figure	Figure 2 in Section 2.3; text description directly below the figure
Environment and dependencies	Python dependencies in requirements.txt; static browser application served through GitHub Pages or a local HTTP server
Run and reproduction steps	python3 scripts/extract_palettes.py --write python3 scripts/extract_palettes.py --check Then use the documented local static-server or GitHub Pages workflow.
Data access or generation steps	Source photographs → scripts/extract_palettes.py → manifest and generated photo/place palette records → browser application
Tests and expected outputs	50 photographs/5 places regenerate; fixed-seed reproducibility; Sharma CIEDE2000 tests pass; offline/browser consistency; self=100; A↔B symmetry; Jinxi↔Bacheng=90/100, weighted ΔE00=10.01
Known issues and fallback	Limits: k=5 is lossy; photographic coverage is small/uneven; small culturally salient accents may be missed. Fallback: documented local static-server/static-browser workflow.
License, citations, privacy, and public-sharing decision	No explicit open-source license is assigned. Public repository access does not imply unrestricted reuse. Photographs remain team-authored material.

Appendix C User Experience Protocol and Evidence

This record summarizes in-class peer evaluation and progress critique, not a formal recruited user study or broad validation.

Table C1. User-experience session record.

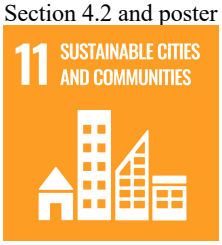
Field	Record
Participant role and consent or notice	In-class design/media student peers and instructor feedback; approximately two verbal comments and five written Ed Discussion comments. No participant identifiers were recorded in the project evidence log.
Exact task and starting state	During classroom review, explore the default atlas; select a place/photo; inspect source-linked palettes and similarity; compare samples; use the Compare interaction where relevant; discuss evidence limits.
Success criterion and measures	Intended design goals: understand sampled-palette similarity as distinct from cultural similarity, trace results to source photographs, and recover/reset. Not every reviewer was observed completing every step.
Actions, hesitation, errors, and recovery	Observed issues: algorithm difficult to explain/defend; advanced effect missing; earlier navigation/link visibility concerns; temporal/seasonal comparison suggested.
System response, evidence inspected, and task outcome	Reviewers inspected the prototype, palettes, similarity views, and comparison flow; feedback led to algorithm, Compare-effect, documentation, and UI revisions.
Accessibility or interpretation barrier	Back-button visibility; dashed-link visibility; comparison workflow and advanced-effect discoverability may still benefit from clearer in-context instructions.
Observed strength, priority, and question	Strength: source-linked comparison and place exploration supported specific critique. Priorities: algorithm grounding, advanced effect, and readability. Scope question: seasonal/time/weather variation.
Feedback decision and visible revision	Revise algorithm; Revise/add advanced Compare; Revise visibility; Investigate temporal extension.
Evidence boundary and next validation	Classroom critique only: no broad usability, learning, cultural validity, community-benefit, or SDG-impact claims. Next: broader intended-user testing plus algorithm/human-similarity validation.

Privacy and evidence handling

Peer identities were visible within the course environment but were not recorded in the project evidence log. Only task/design-relevant feedback was retained on the three team members' laptops; no identifiable peer feedback is included in public materials.

Appendix D Feedback and Response Log

Table D1. Progress activities and Week 7 duet feedback-to-revision record.

Activity or ID	Evidence received	Keep, Revise, or Investigate and reason	Change or next test and owner	Artifact or commit
Interactive Demonstration (Sep 24)	Back button too small; dashed links too light; seasonal/time-of-day comparison suggested	Revise visibility — completed; Investigate temporal extension — not prioritized in current scope	Increased button prominence and link contrast; documented temporal conditions as sampling limits. Owners: Interaction & Evaluation + Data Visualization & Prototyping.	Current artifact; commit ebe8720
Peer Evaluation (Sep 29)	Algorithm grounding unclear; advanced effect missing; temporal feature suggested; interaction/readability discussed	Revise algorithm; Revise advanced effect; Revise readability; Investigate temporal extension	Implemented CIELAB/K-means++/CIEDE2000 pipeline, Compare effect, reproducibility records, and UI fixes. Owners: Data Visualization & Prototyping + Interaction & Evaluation.	Commit 1979de7
Response to Peers	Team reviewed peer/instructor comments and course requirements	Implemented highest-priority revisions; retained temporal feature as out-of-scope investigation	Algorithm + reproducibility + advanced Compare + UI visibility; temporal limitation documented	Current artifact / repository / 754758f
Week 7 duet question and defense (Oct 13)	Jiajun Fu, Ruojin Wang, and Enwei Fan asked us to clarify the project motivation and specific community contribution.	Keep / Clarify — the question did not require an artifact change, but clarified the motivation, intended creative audience, and bounded SDG contribution.	Clarify final communication: source-linked visual references can help creators form intuitive impressions of place-based color; broader cultural communication and SDG 11.4 relevance remain intended, not demonstrated. Owner: Team.	No artifact change from defense; final paper/poster communication revision.
Figure 1: five actual, legible screenshots + evidence and limits				
Official SDG icon, exact target, bounded contribution	Section 4.2 and poster SDG panel; 			
Full acknowledgements + separate AI disclosure	Main-report acknowledgements and separate AI-and-human-responsibility section; reviewer contributions specified.			
Full direct URLs for artifact, repository, poster, paper and video	Artifact/repository/demo available; https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Poster.pdf; https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Report.pdf; 4368799			

Feedback is credited while design authorship remains with the team; recommendations with insufficient evidence are marked Investigate and paired with a next test.

Appendix E Poster and Demo Evidence

Table E1. Poster preflight.

Check	Evidence or location
Descriptive title; evidence-supported claim and adjacent boundary	Available in the revised report; claim and boundary remain adjacent.
Question, users, exact task, verified data, and five representations	Content is synchronized with the submitted artifact.
Repository and software architecture diagram from Figure 2	Completed as Figure 2 in Section 2.3, with repository paths, data flow, checks, deployment, and fallback.

Check	Evidence or location
User-experience walkthrough and revision diagram from Figure 3	Walkthrough/revision evidence is documented.
Four validation levels and all three frontiers, with one primary	Report marks Human-Centered Spatiotemporal Intelligence as primary.
Local-community service and exact SDG goal and target	Report states SDG 11: Sustainable Cities and Communities, Target 11.4 with a bounded contribution.
Artifact/repository/poster/paper/video URLs, team roles, provenance, accessibility limits, and textual figure descriptions are included.	Provenance, roles, fallback, and accessibility limits are documented.

Table E2. Demo storyboard and transcript map.

Embeded Demo video Directly on the reproducible repository:

https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_DemoVideo.mp4

Time	Video segment	Required content and evidence
0:00-0:15	Opening	Project title, research question, intended users, and comparison goal across sampled color palettes from places in China.
0:15-0:35	Method	Research-grounded color extraction method, with the clearer algorithm explanation and Method-section illustration.
0:36-1:16	Geographic atlas and source views	Homepage geographic visualization of five sampled places; select a location to inspect its source photographs and recorded color palettes.
1:17-1:47	Comparison and advanced interaction	Draggable divider with continuously updated weighted blend palette; static comparison fallback; A/B comparison of photos or places; network idiom showing similarities and differences in recorded colors.
1:48-2:07	User evidence and revision	Peer/instructor review led to algorithm revision and a more visual, understandable network idiom; revisions support comparison, traceability, recovery, and Human-Centered Spatiotemporal Intelligence.
2:08-2:57	Contribution, limitations, and next validation	SDG 11.4 alignment; small/uneven sample and five-color-summary limitations; next testing with intended users, human–algorithm similarity comparison, and contextual feedback from residents/cultural practitioners.

Table E3. Professional reference artifacts.

Reference	What to study	Course extension
CHI 2026 Interactive Demos: https://chi2026.acm.org/authors/interactive-demos/	Self-contained demo paper, walkthrough video, accessible submission, and successful examples.	Add both required diagrams, the three-frontier contribution, local service, and exact SDG target.
Zhou et al. TactorBots demo paper: https://doi.org/10.1145/3544549.3583897	How a short demo paper explains a working system and the interaction experience.	Make repository structure, data provenance, evidence boundaries, and correction routes explicit.
TactorBots video: https://www.youtube.com/watch?v=mGvSld62Mhk	How the system, user flow, and physical or interface behavior become visible on video.	Keep the course video to 2:59 or shorter and include the Sep 29 evidence and revision.
Zhou et al. Shape-Kit demo paper: https://doi.org/10.1145/3706599.3721280	How a recent CHI demonstration connects system design, interaction, and a concise research contribution.	Add the repository architecture, the task-based walkthrough, and explicit evidence boundaries.
Shape-Kit video: https://www.youtube.com/watch?v=JSUDrGJjpWw	How a working system and user experience can be communicated through a short demonstration.	Use the stricter course duration and show the feedback-to-revision trace.
CHI 2026 Posters: https://chi2026.acm.org/authors/posters/	Title, concise overview, clear illustrations, visual legibility, and links to online materials.	Show software architecture, UX walkthrough, three capacities, local community, and SDG accountability.
AAAI-26 Student Abstract and Poster Program: https://aaai.org/conference/aaai/aaai-26/student-abstract-and-poster-program-call/	Concise paper-to-poster translation in a major AI conference context.	Preserve the visualization validation and public-service requirements of this course.
SIGCHI Guide for Videos: https://sigchi.org/resources/guides-for-authors/videos/	Captions, audio, readability, permissions, encoding, and accessible visual description.	Follow the course's 2:59 maximum and verify every caption and technical term.

Appendix F Team Contribution and AI Assistance

Table F1. Named team contributions.

Member	Roles and concrete contributions	Evidence or commit	Design element the member can defend
Andi Wan	Data Visualization and Interaction Design: network and spatial idioms, expanded color comparison visualizations, and refinement of the prototype's comparative data presentation.	Commit 1979de7	Visualization idioms, comparative encoding, spatial/network interaction, and visualization design rationale.
Jiaxin Wang	Prototype Framework and Data Curation: core prototype structure, initial image selection/curation, color palette extraction, basic place comparison, and foundational interaction logic.	Commit ebe8720	Prototype structure, data curation, core interaction flow, and basic comparison design rationale.
Tingyi Xiao	Color Processing and Advanced Comparison: color extraction algorithm refinement, comparison logic, and implementation of the Advanced Effects "Comparison" feature.	Commit 754758f	Color extraction algorithm, palette processing, comparison logic, and advanced effect rationale.

Material AI use is disclosed below; the team reviewed outputs and retained responsibility for all submitted claims, design decisions, code, and evidence.

Table F2. AI assistance and human verification.

Field	Disclosure
Tool, displayed model or version, and date	ChatGPT — exact displayed model/date: GPT 5.6, October 15. Codex — exact displayed model/date: GPT 5.6-Sol Medium, October 15.
Purpose and significant prompt or input summary	ChatGPT: project framing, report planning, language drafting, evidence-boundary wording, and appendix organization. Codex: interface implementation, code generation/debugging, repository-based technical support, and document formatting.
Output accepted and where it appears	Selected wording/structure and code or document edits reviewed by the team; accepted material appears in the report, appendices, and implementation where applicable.
Output rejected and why	Unsupported or speculative claims, unsupported implementation details, and incorrect code must be excluded or removed.
Human changes to argument, visuals, data, or code	The team selected the question, verified evidence, revised wording, reviewed/tested code, and retained responsibility for design, interpretation, and claims.
Original sources, data, code, and outputs checked	Official assignment template, project files, repository, project data, and team evidence.
Failures corrected and remaining unverified points	Remaining limitations concern broader user validation, parameter sensitivity, human-perception comparison, and contextual/community interpretation.

Team responsibility statement

We designed and verified the submitted work, distinguished evidence from inference, credited reused resources and contributions, protected private or restricted material, and can explain the project's question, data, design, implementation, evidence, limitations, local-community contribution, and SDG connection.

Team member signatures or typed names: Andi Wan, Tingyi Xiao, Jiaxin Wang Date: October 15, 2026